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Posets

Partially Ordered Relation and Poset:-

A relation R on set A is called partial order if R is reflexive, antisymmetric and transitive.

The set A together with the partial ordered R is called a partially ordered set or simply a poset. It is denoted by (A, R) or (A, ≤)

Example:- Define the relation R on set (z) by aRb if a-b is non-negative even integer. Verify that R defines a partial order for z.

→ R is reflexive, if aRa

for aRa, the (a-b) becomes (a-a)

a-a = 0 ---- is a non-negative even integer.

∴ R is reflexive.

R is not symmetric, since as aRb then b$\not R$a

for aRb, a-b = 2n where n is integer number.

then for bRa, b-a = -2n which is ~~not even~~ negative even number.

R is transitive since,

let aRb, a-b = $2n_1$

& bRc, b-c = $2n_2$

then aRc, a-c = (a-b)+(b-c)

= $2n_1 + 2n_2 = 2(n_1+n_2)$ ---- which is non-negative even no.

R is antisymmetric, since [aRb & bRa then a=b]

If a-b = 0 & b-a = 0 OR [a$\not R$b & b$\not R$a then a≠b]

then a=b

It is reflexive, antisymmetric and transitive, hence given relation is partial order relation.

Hasse Diagram:- It is a graphical representation of a partial ordering relation in which all arrowheads are to be pointing upward is known as Hasse diagram of the relation.

Procedure of drawing Hasse diagram:-

1. Draw the digraph of the given relation.
2. Delete all ~~cycle~~ loops of vertices from digraph.
3. Eliminate all edges that are implied by the transitive relation. for e.g. If aRb, bRC then aRC. So eliminate (a,c) edges (arrows)
4. Draw the digraph of a partial order with all edges pointing upward so that arrow may be omitted from edges.
5. finally replace the circles representing the vertices by dots.

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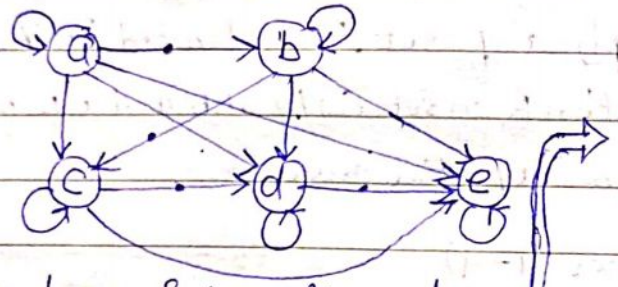
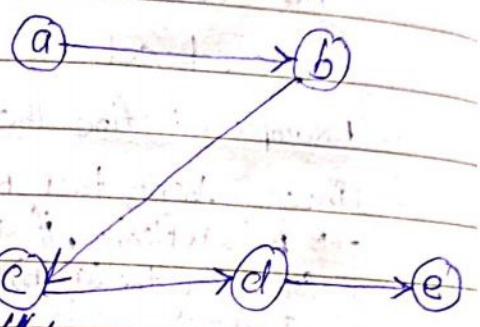
Ex:- Draw Hasse diagram of following relation. What diagram is called as?
Let $A = \{a, b, c, d, e\}$ and $R = \{(a,a), (b,b), (c,c), (d,d), (e,e), (a,b), (b,c), (c,d), (d,e), (a,c), (a,d), (a,e), (b,d), (b,e), (c,e)\}$

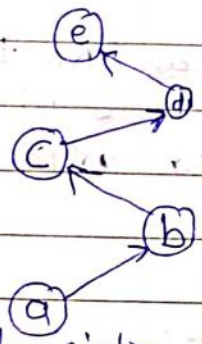
→ ① Digraph of the relation R is shown below.



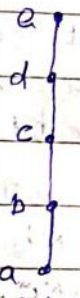


② Remove loops & transitive edges

$(a,b), (b,c), (\cancel{a,c})$
$(a,b), (b,e), (\cancel{a,e})$
$(b,c), (c,d), (\cancel{b,d})$
$(b,d), (d,e), (\cancel{b,e})$
$(c,d), (d,e), (\cancel{c,e})$
$(a,b), (b,d), (\cancel{a,d})$



③ All edges pointing upword



④ replace circles by dots and remove arrows from edges.

This Hasse diagram is a chain. because every two elements are related

$(A, \leq)$ be a partially ordered set (Poset) is called <u>'chain'</u> if every two elements in the subset are related. The number of elements in a chain is called the length of chain.

A partially ordered set $(A, \leq)$ is called a "totally ordered set" if it is a chain.

$(A, \leq)$ be a Poset called an <u>'antichain'</u> if any two elements in the subset are not related.

Ex - Draw the Hasse diagram of the following sets under the partial ordering relation divides and indicate those which are chains.

(i) $A = \{2, 4, 12, 24\}$, (ii) $A = \{1, 3, 5, 15, 30\}$

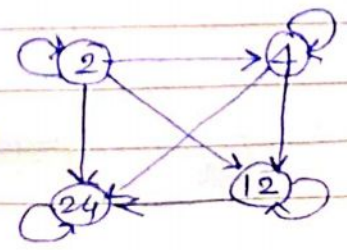
→ (i) $A = \{2, 4, 12, 24\}$. Partial ordering divisibility onset A is as follows.

$R = \{(2,2), (2,4), (2,12), (2,24), (4,4), (4,12), (4,24), (12,12), (12,24), (24,24)\}$

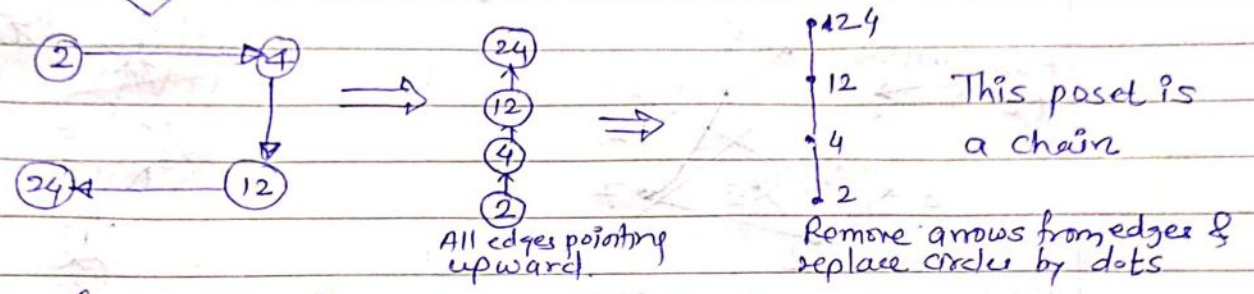
then matrix of R is

$$\begin{array}{c|cccc} & 2 & 4 & 12 & 24 \\ \hline 2 & 1 & 1 & 1 & 1 \\ 4 & 0 & 1 & 1 & 1 \\ 12 & 0 & 0 & 1 & 1 \\ 24 & 0 & 0 & 0 & 1 \end{array}$$

Digraph $\xrightarrow{\text{To}}$ (Hasse diagram)



Remove loops and transitive edges [(2,12), (2,24), (4,24)]

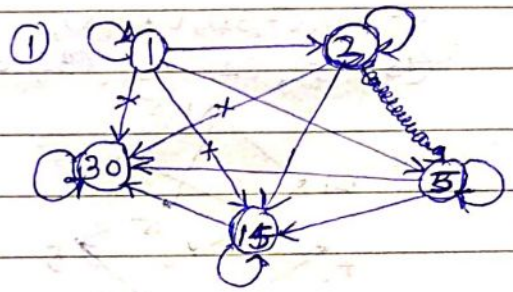


(ii) $A = \{1, 3, 5, 15, 30\}$

→ $R = \{(1,1), (1,3), (1,5), (1,15), (1,30), (3,3), (3,15), (3,30), (5,5), (5,15), (5,30), (15,15), (15,30), (30,30)\}$

$$\begin{array}{c|ccccc} & 1 & 3 & 5 & 15 & 30 \\ \hline 1 & 1 & 1 & 1 & 1 & 1 \\ 3 & 0 & 1 & 0 & 1 & 1 \\ 5 & 0 & 0 & 1 & 1 & 1 \\ 15 & 0 & 0 & 0 & 1 & 1 \\ 30 & 0 & 0 & 0 & 0 & 1 \end{array}$$

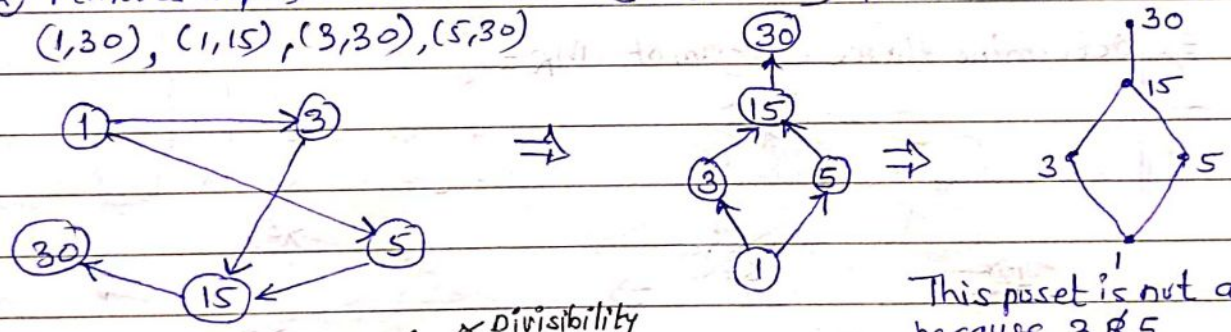
⇒ Digraph



② Removes loops & transitive edges (1,30), (1,15), (3,30), (5,30)

③ Edges pointing upward

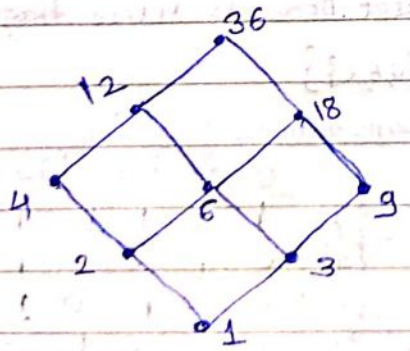
④ Hasse diagram



This poset is not a chain because 3 ∤ 5

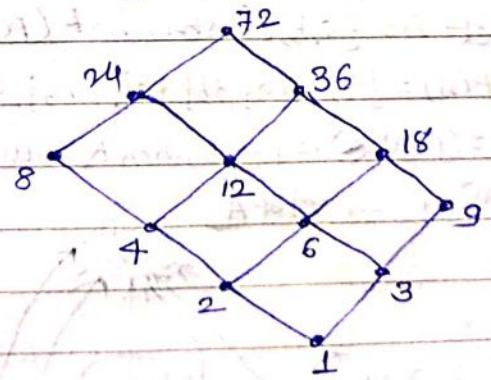
Ex Draw Hasse diagram of D36 (Divisibility)

→ $D36 = \{1, 2, 3, 4, 6, 9, 12, 18, 36\}$



D72:-

$D72 = \{1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72\}$



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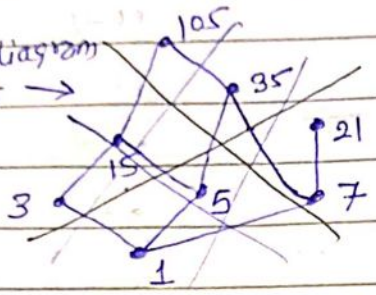
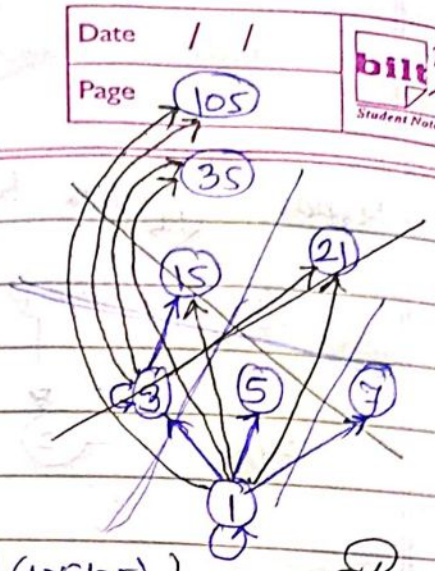
Ex Draw Hasse digram of D105

→ D105 = {1, 3, 5, 7, 15, 21, 35, 105}

Hasse diagram of D105 →

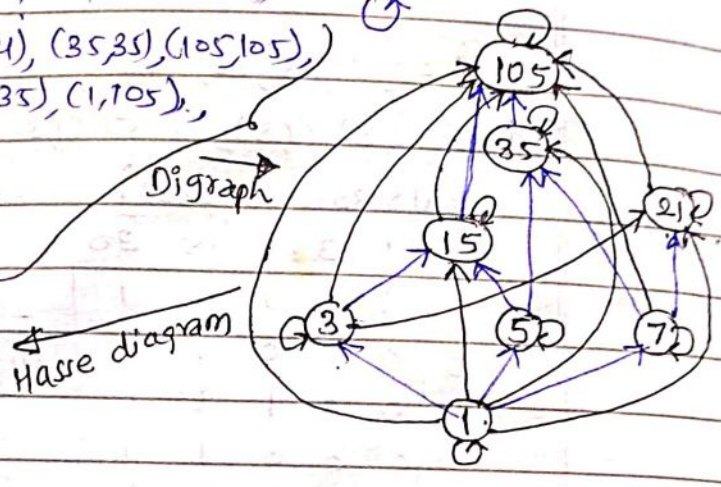


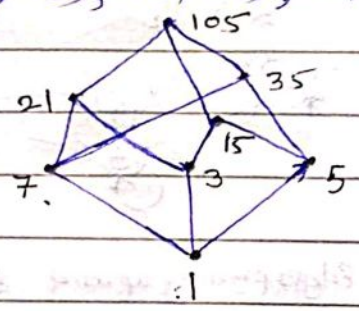


R = { (1,1), (3,3), (5,5), (7,7), (15,15), (21,21), (35,35), (105,105),
(1,3), (1,5), (1,7), (1,15), (1,21), (1,35), (1,105),
(3,15), (3,21), (3,105),
(7,21), (7,35), (7,105),
(15,105), (21,105), (35,105)}

Digraph

Hasse diagram

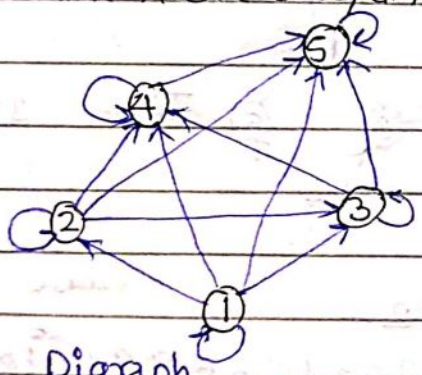




Ex Determine Hasse diagram of

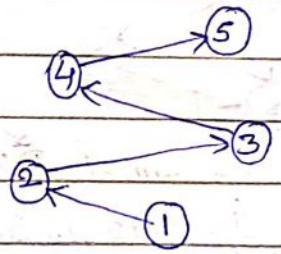
$$M_R = \begin{array}{c|ccccc} & 1 & 2 & 3 & 4 & 5 \\ \hline 1 & 1 & 1 & 1 & 1 & 1 \\ 2 & 0 & 1 & 1 & 1 & 1 \\ 3 & 0 & 0 & 1 & 1 & 1 \\ 4 & 0 & 0 & 0 & 1 & 1 \\ 5 & 0 & 0 & 0 & 0 & 1 \end{array}$$

→



Digraph

⇒



⇒

Hasse diagram

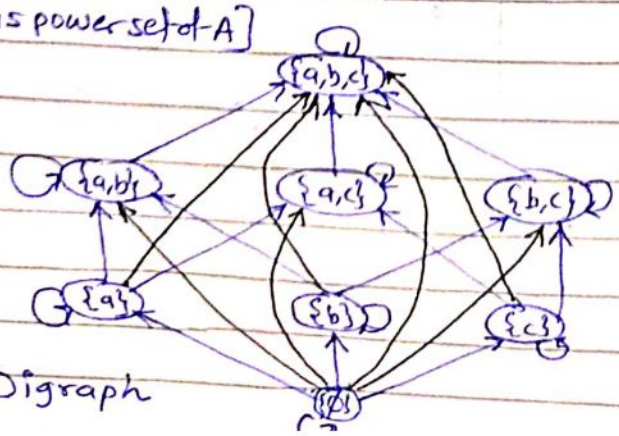
This poset is a cha

Ex Let A = {a, b, c}. show that (P(A), ≤) is a poset. Draw its Hasse diagram.

→ P(A) = { {ϕ}, {a}, {b}, {c}, {a,b}, {a,c}, {b,c}, {a,b,c} }

A = (P(A), ≤) is poset, then R must be reflexive, antisymmetric & transitive.

[P(A) is power set of A]

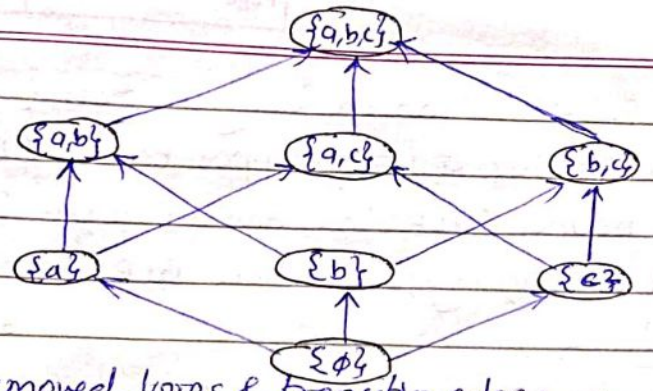


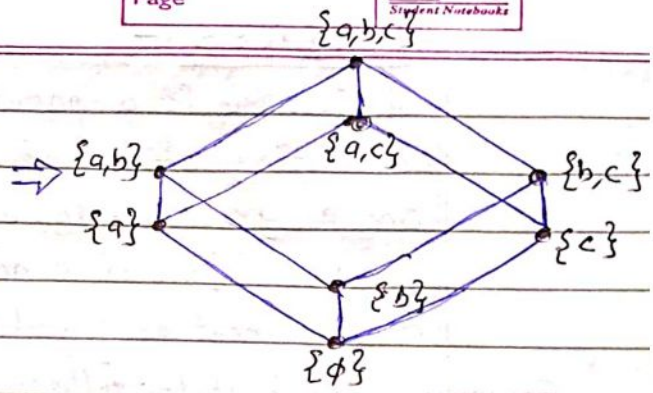
Digraph

$$M_R = \begin{array}{c|cccccccc} & \{\phi\} & \{a\} & \{b\} & \{c\} & \{a,b\} & \{a,c\} & \{b,c\} & \{a,b,c\} \\ \hline \{\phi\} & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ \{a\} & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 \\ \{b\} & 0 & 0 & 1 & 0 & 1 & 0 & 1 & 1 \\ \{c\} & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ \{a,b\} & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ \{a,c\} & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ \{b,c\} & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ \{a,b,c\} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{array}$$

Matrix

(Put '1' where row element is divisor of column)



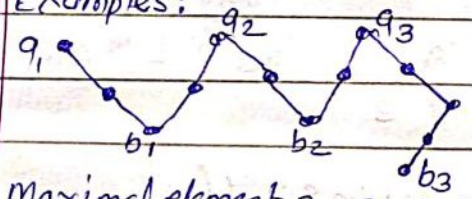


Removed loops & transitive edges

Maximal Element:- An element 'a' ($a \in A$) is called maximal element of A if there is no element 'c' in A such that $a < c$

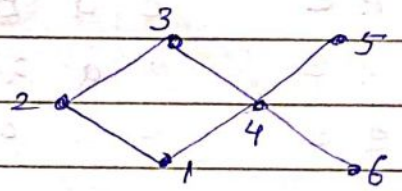
Minimal Element:- An element 'b' ($b \in A$) is called minimal element of A if there is no element 'c' in A such that $c < b$

Examples:



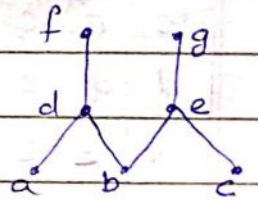
Maximal elements = a_1, a_2, a_3

minimal elements = b_1, b_2, b_3



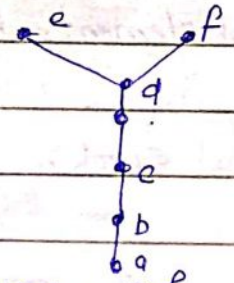
Max = 3, 5

Min = 1, 6



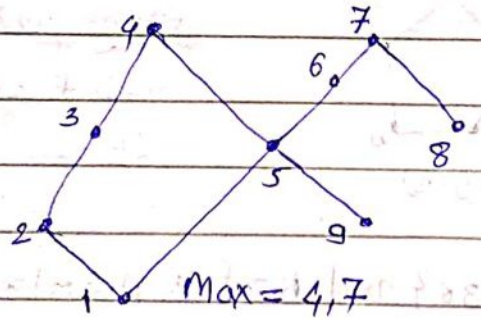
Max. Element = f, g

Min. Element = a, b, c



Max. = e, f

Min = a



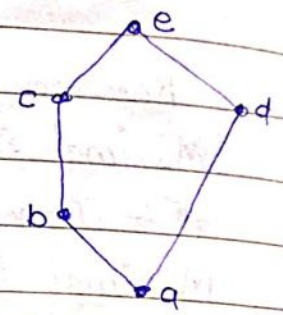
Max = 4, 7

Min = 1, 9, 8

Lattices:-

A Lattice is a poset $(L, \leq)$ in which every subset $\{a,b\}$ consisting of two elements has a least upper bound (LUB) and a greatest lower bound (GLB). We denote LUB $(\{a,b\})$ by $a \vee b$ (join of a and b). similary, we denote GLB $(\{a,b\})$ by $a \wedge b$ (meet of a and b).

Example:- Check if the following diagram is lattice or not



→ LUB :

$\vee$	a	b	c	d	e
a	a	b	c	d	e
b	b	b	c	e	e
c	c	c	c	e	e
d	d	e	e	d	e
e	e	e	e	e	e

GLB :-

$\wedge$	a	b	c	d	e
a	a	a	a	a	a
b	a	b	b	a	b
c	a	b	c	a	c
d	a	a	a	d	d
e	a	b	c	d	e

This is lattice because every pair of elements has a LUB and GLB.

Example:- Which of the figures (a) & (b) represents lattice & justify

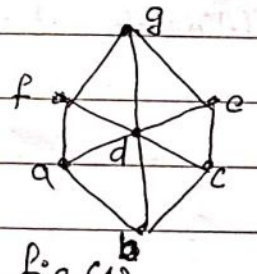
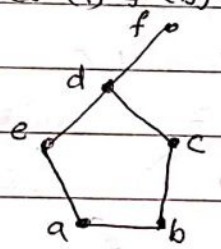


fig.(1)



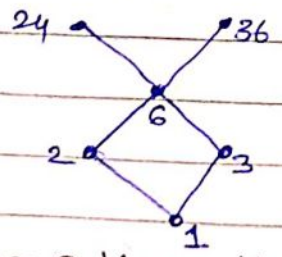
fig(2)

Sol^n:- fig(a) & (b) are lattices as every pair of elements has GLB & LUB. (Prepare LUB & GLB table and check)

Ex:- Let $X = \{1,2,3,6,24,36\}$ and $R = \{(x,y) \in R \mid x \text{ divides } y\}$ $\quad x \text{ divides } y = \frac{y}{x}$

→ $R = \{(1,1),(1,2),(1,3),(1,6),(1,24),(1,36),(2,2),(2,6),(2,24),(2,36),(3,3),(3,6),(3,24),(3,36),(6,6),(6,24),(6,36),(24,24),(36,36)\}$

Hasse diagram:-



Maximal Element = 24,36

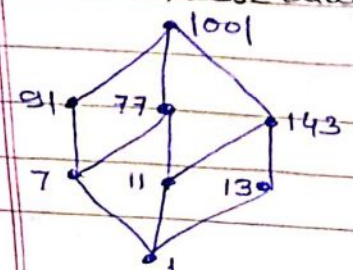
Minimal element = 1

chain = $\{(2,6,36),(3,6,24)\}$

Antichain = $\{(2,3),(24,36)\}$

This poset is not a lattice as $24 \vee 36 = \phi$

Ex:- Draw Hasse diagram of D1001. $D_{1001} = \{1,7,11,13,77,91,143,1001\}$



It is distributed Lattice

Inverse of 91 is 143

" of 143 is 91

" 7 is 13

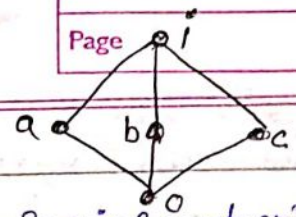
" 13 is 7

" 77 is 13

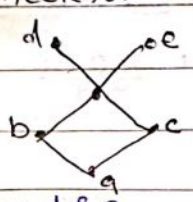
Inverse of 11 is 91

1001
143×7
91×11
77×13

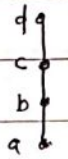
Ex:- Define distributed lattice. check the diagram is distributed lattice or not



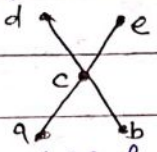
Ex:- check for lattice or not (If for each pair there is join & meet exist is called lattice)



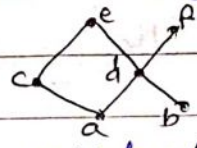
for d & e join not exist
✓ Not lattice



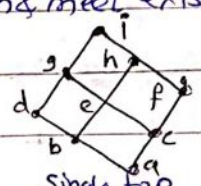
single end
single top
✓ lattice



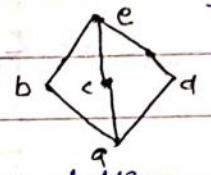
No join for d, e
No meet for a, b
✓ Not lattice



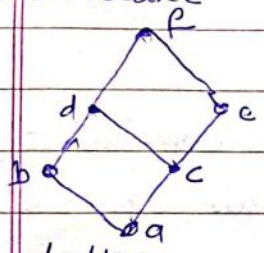
No join for e, f
No meet for a, b
✓ Not lattice



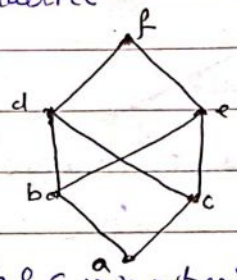
Single top
Single bottom
lattice



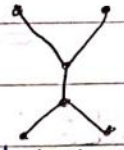
✓ lattice



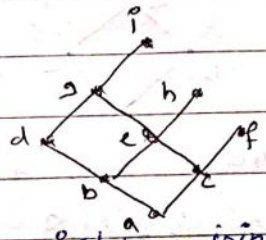
✓ lattice



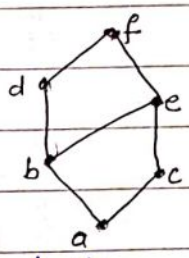
b & c join not exist
d & e has too meet
✓ Not lattice



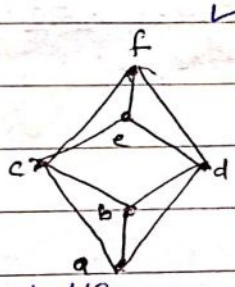
Not lattice



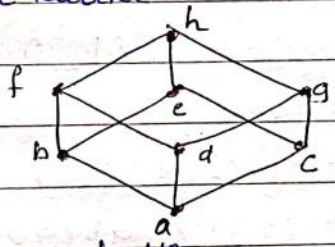
i & h has no join
Not single top
✓ Not lattice



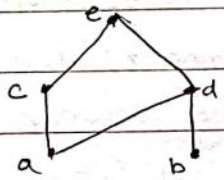
✓ lattice



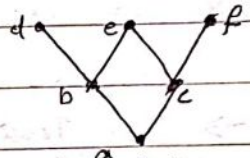
✓ lattice



✓ Lattice



✓ Not lattice



✓ Not lattice

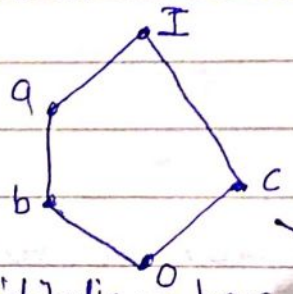
Types of Lattices :- ① Bounded Lattice ② Distributive lattice ③ Complemented lattice

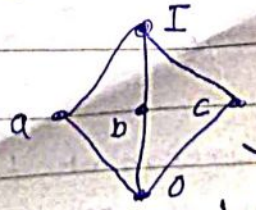
Distributed lattice :- A lattice L is called distributive if for any elements a, b and c in L then we have the following distributive properties.

1. $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$
2. $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$

If L is not distributive, we say that L is non-distributive.

Ex:- show that the lattice in below figures are non-distributive





Sol^n → for distributive, we have

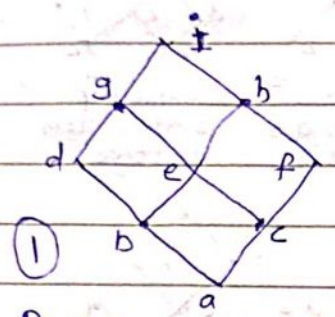
$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$
$a \wedge (I) = (b) \vee (o)$
$\therefore a = b$
$\therefore$ is non-distributive

$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge$
$a \wedge (I) = (o) \vee (o)$
$a = o$
$\therefore$ is non-distributive.

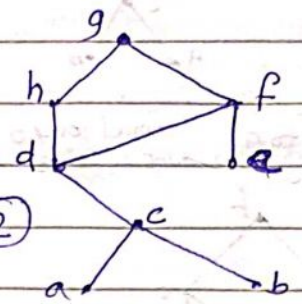
Date / /
Page

Ex:- Draw Hasse diagram of D30 & D20. Draw Hasse diagram of set of integers which divide 24 with 'divisibility' (i.e D24). Draw Hasse of D42 & D70

Ex:- Whether the following Hasse diagram represent lattice?



(1)



(2)

Draw LUB & GLB table (1) Yes, it is lattice as every pair of elements has a GLB & LUB. (2) No, it is not a lattice as pair (a,b), (e,c), (e,d) & (h,e) do not have GLB.

Ex:- Consider the chains of divisors of 4 and 9 i.e $L_1 = \{1,2,4\}$ & $L_2 = \{1,3,9\}$. find partial ordering relation of division on L_1 & L_2. Draw lattice of $L_1 \times L_2$

→ Partially ordering relation of division on,

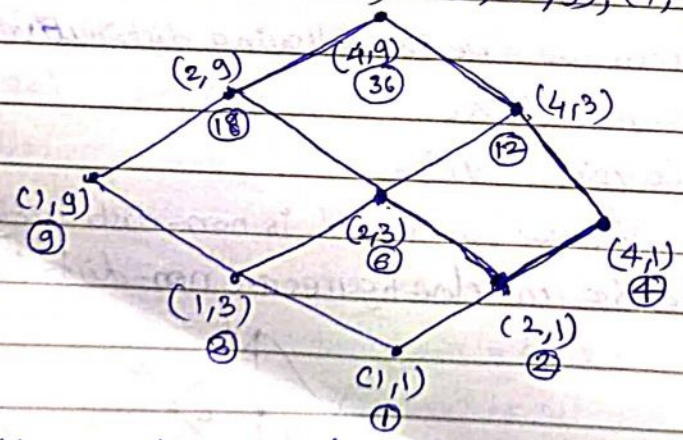
L_1 is $R_1 = \{(1,1), (1,2), (1,4), (2,2), (2,4), (4,4)\}$

L_2 is $R_2 = \{(1,1), (1,3), (1,9), (3,3), (3,9), (9,9)\}$

$L_1 \times L_2 = \{1,2,4\} \times \{1,3,9\}$

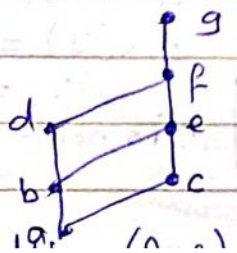
$\therefore \quad = \{\underset{1}{(1,1)}, \underset{3}{(1,3)}, \underset{9}{(1,9)}, \underset{2}{(2,1)}, \underset{6}{(2,3)}, \underset{18}{(2,9)}, \underset{4}{(4,1)}, \underset{12}{(4,3)}, \underset{36}{(4,9)}\}$

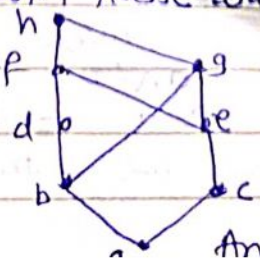
$\therefore \quad = \{\overset{1}{(1,1)}, \overset{2}{(2,1)}, \overset{3}{(1,3)}, \overset{4}{(4,1)}, \overset{6}{(2,3)}, \overset{9}{(1,9)}, \overset{12}{(4,3)}, \overset{18}{(2,9)}, \overset{36}{(4,9)}\}$



Ex. Draw Hasse diagram of D60.

Ex. Determine whether the posets with Hasse diagram are lattice or not (Write LUB & GLB table and verify)





Soln

Ans:- not lattice.